User Scenarios List

short description of all the user scenarios

Nr	Role	Name	Description
US_01	Administrator	Add book to catalog	Administrator: Adds a new book to the library catalog so that users can borrow it.
US_02	Administrator	Edit book information	Administrator: Edits existing book details so that incorrect details can be updated.
US_03	Administrator	Delete book from catalog	Administrator: Removes an outdated book from the catalog so that it is no longer available.
US_04	User	View book details	User: Views detailed information about a book so that they can learn more about it.
US_05	Administrator	Upload book cover	Administrator: Uploads a cover image for a book so that books are visually identifiable.
US_06	Administrator	Add author name	Administrator: Adds the author name to a book so that books are properly categorized.
US_07	Administrator	Add publication year	Administrator: Adds the publication year so that users know when the book was published.
US_08	Administrator	Add book description	Administrator: Adds a description so that users can understand the book content.
US_09	Administrator	Manage number of copies	Administrator: Sets and updates the number of copies so that availability can be tracked.
US_10	User	View list of available books	User: Views the list of available books so that they can choose a book to borrow.

User Scenarios Extended

US_01: Add book to catalog:

1. Administrator navigates to the Book Management section.
2. Administrator clicks 'Add Book' and fills in the form (title, author, ISBN, copies).
3. System validates the input fields.
4. System saves the new book record to the catalog.
5. New book appears in the catalog and is available for borrowing.
6. If ISBN already exists: System displays a duplicate warning.
7. If required fields are missing: Form highlights empty fields.

US_02: Edit book information:

8. Administrator searches for the book in the catalog.
9. Administrator clicks 'Edit' on the selected book.
10. Administrator modifies the desired fields.
11. System validates the updated data.
12. System saves the changes and shows a confirmation message.
13. If book has active loans: System warns that changes apply to future borrows.

US_03: Delete book from catalog:

14. Administrator searches for the book to be removed.
15. Administrator clicks 'Delete' on the selected book.
16. System checks for active loans on this book.
17. If no active loans: System prompts for confirmation.
18. Administrator confirms the deletion.
19. System sets IsActive = false (soft delete).
20. Book no longer appears in the public catalog.
21. If book has active loans: System prevents deletion until all copies are returned.

US_04: View book details:

22. User browses or searches the catalog.
23. User clicks on a book title.
24. System retrieves and displays full book details: title, author, category, ISBN, description, rating, availability.
25. User can read reviews and see current availability status.
26. If book is unavailable: System shows 'No copies available' and offers reservation option.

US_05: Upload book cover:

27. Administrator selects a book from the catalog.
28. Administrator clicks 'Upload Cover' and selects an image file (JPG/PNG).
29. System validates the file format and size (max 2 MB).

30. System stores and links the image to the book record.
31. Cover image is displayed in the catalog.
32. If format is unsupported: Error message shown.
33. If file is too large: Error with size limit shown.

US_06: Add author name:

34. Administrator opens the Add/Edit Book form.
35. Administrator enters the author name or selects from existing authors.
36. System checks if author already exists.
37. System links the author to the book record.
38. Book is displayed with the author name in the catalog.
39. If author does not exist: System creates a new author record automatically.

US_07: Add publication year:

40. Administrator opens the Add/Edit Book form.
41. Administrator enters the publication year.
42. System validates the year (must be a valid 4-digit year, not in the future).
43. System saves the publication year with the book record.
44. Publication year is displayed on the book detail page.
45. If year is invalid: System shows a validation error.

US_08: Add book description:

46. Administrator opens the Add/Edit Book form.
47. Administrator enters a description in the description field.
48. System saves the description with the book record.
49. Description is displayed on the book detail page.
50. If description exceeds maximum length: System shows a character limit error.

US_09: Manage number of copies:

51. Administrator selects a book from the catalog.
52. Administrator updates the TotalCopies field.
53. System checks active loans to validate the new count.
54. System recalculates AvailableCopies based on active loans.
55. System saves the updated copy count.
56. Availability status is updated across the catalog.
57. If new count is less than currently borrowed copies: System prevents the reduction.

US_10: View list of available books:

58. User navigates to the Catalog page.
59. System retrieves all active books with AvailableCopies > 0.
60. System displays books with title, author, category, rating, and availability.
61. User can filter by category or search by title/author.

62. User clicks a book to view full details.
63. If no books are available: System displays 'No books available at this time'.

Use Cases

UC_01: Add book to catalog

UC Name	**UC_01 — Add book to catalog**
Dependency	Administrator must be authenticated
Actors	Primary actor: Administrator Secondary actor: None
Preconditions	Administrator is logged in. Book does not already exist in the catalog.
Description of the Main Sequence	64. Administrator navigates to Book Management section. 1. Administrator clicks 'Add Book' and fills in the form. 1. System validates title, author, ISBN, and copies fields. 1. System saves the new book to the catalog. 1. New book is visible and available for borrowing.
Description of the Alternative Sequence	• ISBN already exists: Duplicate warning shown. • Required fields empty: Form highlights missing fields.
Non functional requirements	Book must be saved within 2 seconds. ISBN must be unique in the system.
Postconditions	New book appears in the catalog and can be borrowed by users.

UC_02: Edit book information

UC Name	**UC_02 — Edit book information**
Dependency	UC_01 — book must exist
Actors	Primary actor: Administrator Secondary actor: None
Preconditions	Administrator is logged in. Book exists in the catalog.
Description of the Main Sequence	65. Administrator searches for the book. 2. Administrator clicks 'Edit'. 2. Administrator modifies fields. 2. System validates the updated data. 2. System saves changes and shows confirmation.
Description of the Alternative Sequence	• Book has active loans: Warning shown — changes apply to future borrows.
Non functional requirements	Changes must be saved within 2 seconds.
Postconditions	Book record is updated with the new information.

UC_03: Delete book from catalog

UC Name	**UC_03 — Delete book from catalog**
Dependency	UC_01 — book must exist
Actors	Primary actor: Administrator Secondary actor: None
Preconditions	Administrator is logged in. Book exists and has no active loans.
Description of the Main Sequence	66. Administrator searches for the book. 3. Administrator clicks 'Delete'. 3. System checks for active loans. 3. System prompts for confirmation. 3. Administrator confirms. 1. System sets IsActive = false (soft delete). 1. Book is removed from the public catalog.
Description of the Alternative Sequence	• Book has active loans: Deletion prevented until all copies are returned. • Administrator cancels: No changes made.
Non functional requirements	Soft delete only — data must be preserved. Action logged for audit.
Postconditions	Book is hidden from the catalog. Data is retained in the database.

UC_04: View book details

UC Name	**UC_04 — View book details**
Dependency	None
Actors	Primary actor: User Secondary actor: None
Preconditions	Book exists and is active in the catalog.
Description of the Main Sequence	67. User browses or searches the catalog. 4. User clicks on a book. 4. System retrieves full book details. 4. Details displayed: title, author, ISBN, description, rating, availability.
Description of the Alternative Sequence	• Book unavailable: 'No copies available' shown with reservation option.
Non functional requirements	Page must load within 1.5 seconds.
Postconditions	User has full information about the book.

UC_05: Upload book cover

UC Name	**UC_05 — Upload book cover**
Dependency	UC_01 — book must exist

Actors	Primary actor: Administrator Secondary actor: File Storage Service
Preconditions	Administrator is logged in. Book exists in the catalog.
Description of the Main Sequence	68. Administrator selects a book. 5. Administrator clicks 'Upload Cover' and selects a JPG/PNG file. 5. System validates format and size (max 2 MB). 5. System stores image and links it to the book. 4. Cover is displayed in the catalog.
Description of the Alternative Sequence	• Unsupported format: Error message shown. • File too large: Error with size limit shown.
Non functional requirements	Upload must complete within 5 seconds. Image resized to standard dimensions.
Postconditions	Book cover image is displayed in the catalog listing and detail page.

UC_06: Add author name

UC Name	**UC_06 — Add author name**
Dependency	UC_01
Actors	Primary actor: Administrator Secondary actor: None
Preconditions	Administrator is logged in. Book form is open.
Description of the Main Sequence	69. Administrator enters author name or selects from existing authors. 6. System checks if author already exists. 6. System links author to the book record. 6. Author name is displayed in the catalog.
Description of the Alternative Sequence	• Author does not exist: System creates a new author record automatically.
Non functional requirements	Author search must return results within 1 second.
Postconditions	Book is properly attributed with author name.

UC_07: Add publication year

UC Name	**UC_07 — Add publication year**
Dependency	UC_01
Actors	Primary actor: Administrator Secondary actor: None
Preconditions	Administrator is logged in. Book form is open.

Description of the Main Sequence	70. Administrator enters the 4-digit publication year. 7. System validates: must be a valid year, not in the future. 7. System saves the year with the book record. 7. Year is displayed on the book detail page.
Description of the Alternative Sequence	• Invalid year: System shows validation error.
Non functional requirements	Validation must be instant (client-side).
Postconditions	Publication year is saved and visible to users.

UC_08: Add book description

UC Name	**UC_08 — Add book description**
Dependency	UC_01
Actors	Primary actor: Administrator Secondary actor: None
Preconditions	Administrator is logged in. Book form is open.
Description of the Main Sequence	71. Administrator enters a description in the text field. 8. System saves the description with the book record. 8. Description is shown on the book detail page.
Description of the Alternative Sequence	• Description exceeds max length: System shows character limit error.
Non functional requirements	Description field must accept at least 1000 characters.
Postconditions	Book description is saved and visible to users on the detail page.

UC_09: Manage number of copies

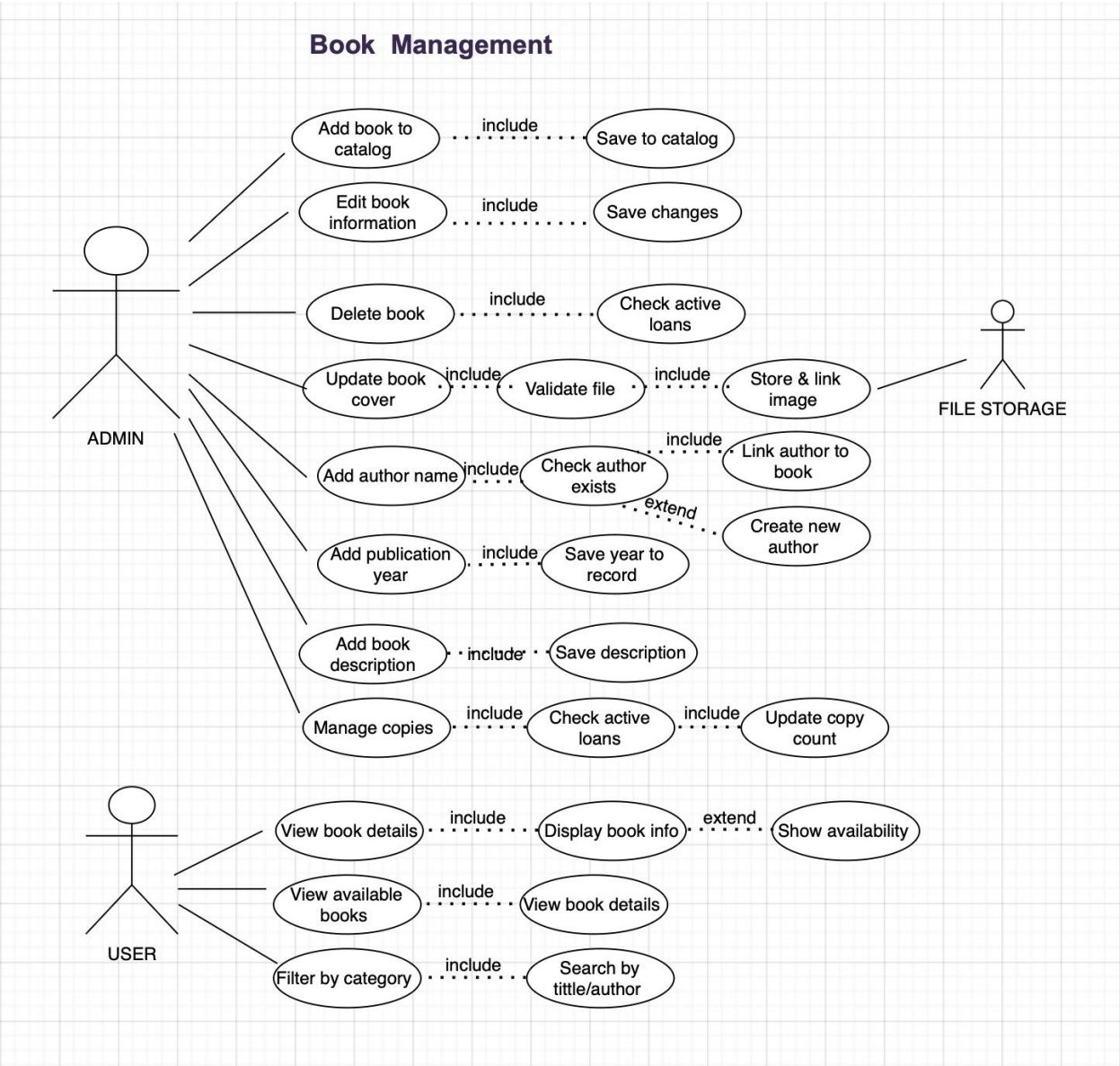
UC Name	**UC_09 — Manage number of copies**
Dependency	UC_01
Actors	Primary actor: Administrator Secondary actor: None
Preconditions	Administrator is logged in. Book exists in the catalog.
Description of the Main Sequence	72. Administrator selects a book. 9. Administrator updates the TotalCopies field. 9. System checks active loans to validate the new count. 8. System recalculates AvailableCopies. 5. System saves updated count. 2. Availability status is updated in the catalog.

Description of the Alternative Sequence	• New count less than borrowed copies: System prevents reduction.
Non functional requirements	Copy count update must be reflected in real-time.
Postconditions	Book availability is accurately updated across the catalog.

UC_10: View list of available books

UC Name	**UC_10 — View list of available books**
Dependency	None
Actors	Primary actor: User Secondary actor: None
Preconditions	At least one book with AvailableCopies > 0 exists.
Description of the Main Sequence	73. User navigates to the Catalog page. 10. System retrieves active books with available copies. 10. Books displayed with title, author, category, rating, availability. 9. User can filter by category or search by title/author. 6. User clicks a book to view full details.
Description of the Alternative Sequence	• No books available: 'No books available at this time' shown.
Non functional requirements	Catalog must load within 2 seconds. Search must be case-insensitive.
Postconditions	User sees the full list of available books and can select one to borrow.

Use Case Diagrams



Functional Requirements

Req#	Role	Requirement	Comments	Priority	Date	Reviewed
FR_01	Administrator	The system shall allow administrators to add new books with title, author, ISBN, category, copies, description, and publication year.	ISBN must be unique. All mandatory fields validated.	1	01/04/2026	Xhim Zotaj
FR_02	Administrator	The system shall allow administrators to edit	Warning shown if	1	01/04/2026	Xhim Zotaj

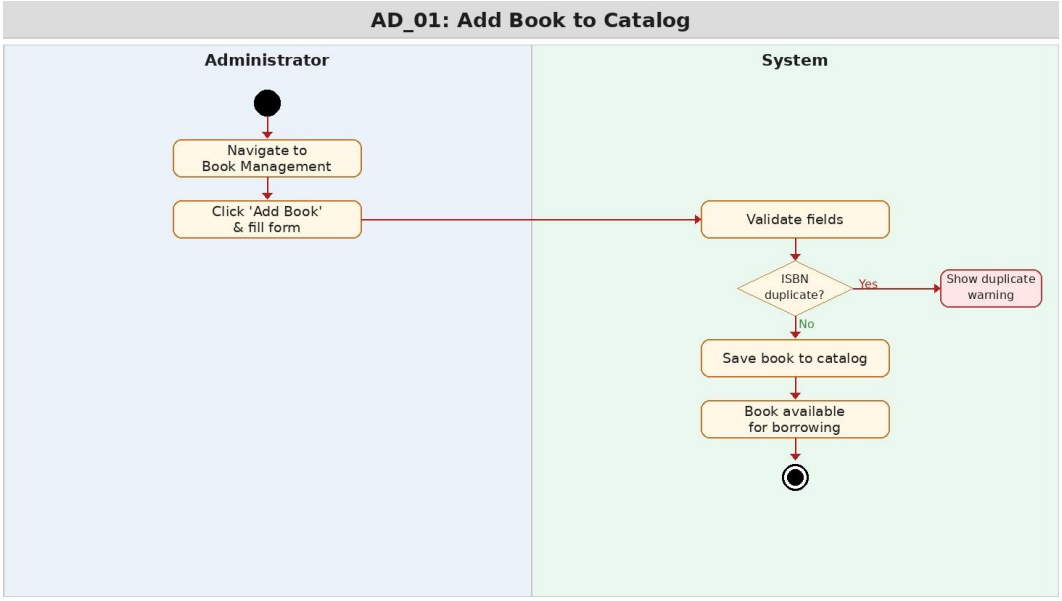
		any field of an existing book record.	book has active loans.			
FR_03	Administrator	The system shall allow administrators to soft-delete books by setting IsActive = false.	Deleted books hidden from catalog. Data preserved in database.	1	01/04/2026	Xhim Zotaj
FR_04	User	The system shall display full book details to users: title, author, ISBN, description, rating, category, and availability.	Page must load within 1.5 seconds.	2	01/04/2026	Xhim Zotaj
FR_05	Administrator	The system shall allow administrators to upload JPG/PNG cover images up to 2 MB for each book.	Unsupported formats or oversized files must be rejected.	2	01/04/2026	Xhim Zotaj
FR_06	Administrator	The system shall allow administrators to assign an author to a book. If the author does not exist, a new record must be created.	Author search returns results within 1 second.	2	01/04/2026	Xhim Zotaj
FR_07	Administrator	The system shall validate the publication year as a 4-digit number not in the future.	Client-side validation required for instant feedback.	3	01/04/2026	Xhim Zotaj
FR_08	Administrator	The system shall allow administrators to add a book description of at least 1000 characters.	Description shown on book detail page.	2	01/04/2026	Xhim Zotaj
FR_09	Administrator	The system shall allow administrators to update TotalCopies. AvailableCopies must be recalculated automatically based on active loans.	Reduction below active loan count must be prevented.	1	01/04/2026	Xhim Zotaj
FR_10	User	The system shall display to users a list of books with AvailableCopies > 0, with search by title/author and filter by category.	Catalog must load within 2 seconds. Search must be case-insensitive.	1	01/04/2026	Xhim Zotaj

Non-Functional Requirements

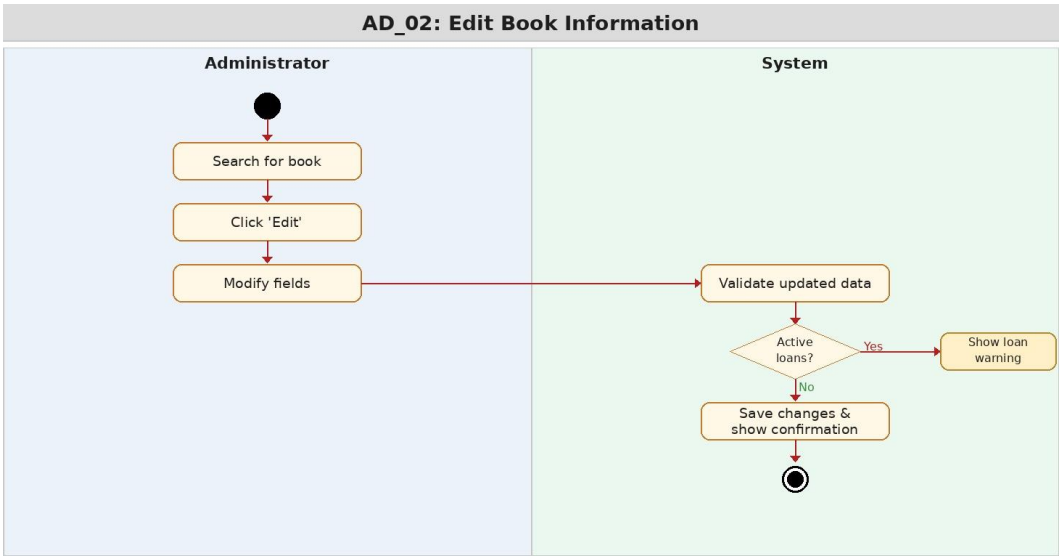
Req#	Category	Requirement
NFR_01	Performance	All book operations (add, edit, view) must respond within 2 seconds under normal load.
NFR_02	Performance	The catalog page must load and render the full book list within 2 seconds for up to 1000 books.
NFR_03	Security	Only authenticated administrators may add, edit, delete, or upload covers for books. Unauthorized requests must return 403 Forbidden.
NFR_04	Security	Soft delete must be used for all book removals. Hard deletes are not permitted to preserve audit history.
NFR_05	Usability	The book add/edit form must provide inline validation feedback without a full page reload.
NFR_06	Usability	The catalog page must support filtering by category and keyword search with results updating in real-time.
NFR_07	Reliability	Book availability counts must be accurate at all times. Race conditions when multiple users borrow the same book must be prevented.
NFR_08	Scalability	The catalog must support at least 10,000 book records and 200 concurrent users without performance degradation.
NFR_09	Maintainability	Book images must be stored separately from the database to keep the database lightweight.
NFR_10	Compliance	ISBN format must conform to ISBN-10 or ISBN-13 standards. Invalid ISBNs must be rejected at input.

Activity Diagrams

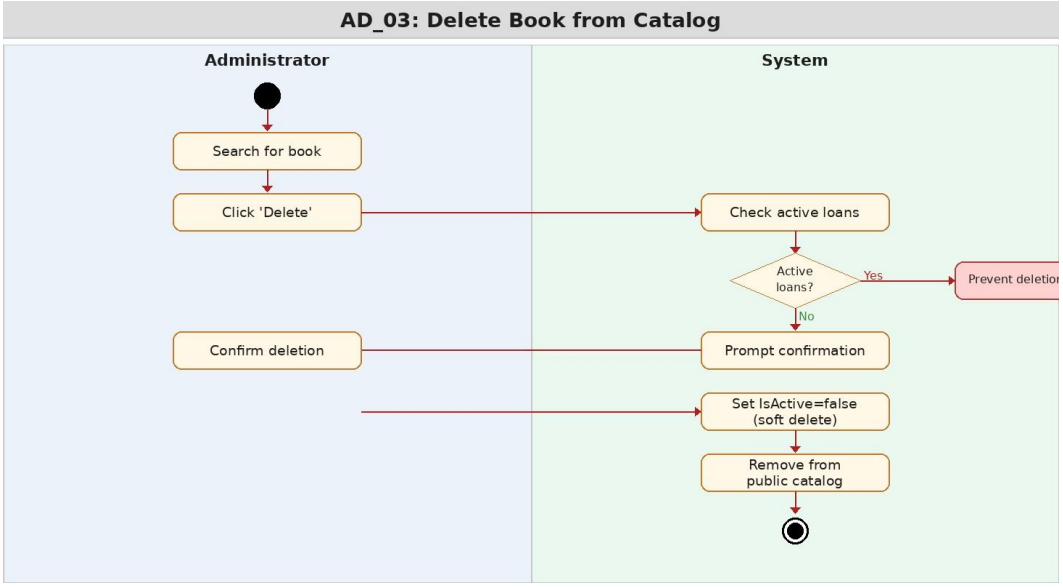
AD_01: Add Book to Catalog



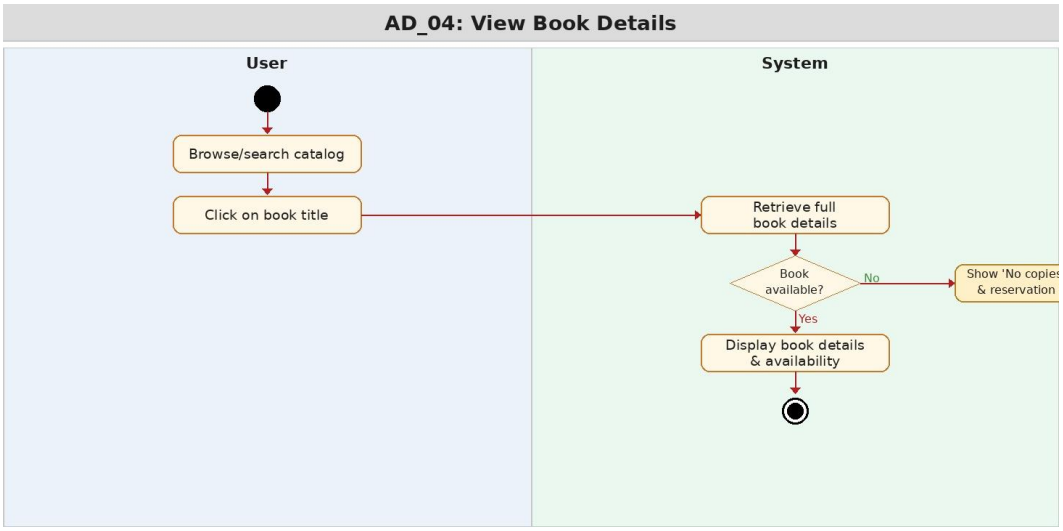
AD_02: Edit Book Information



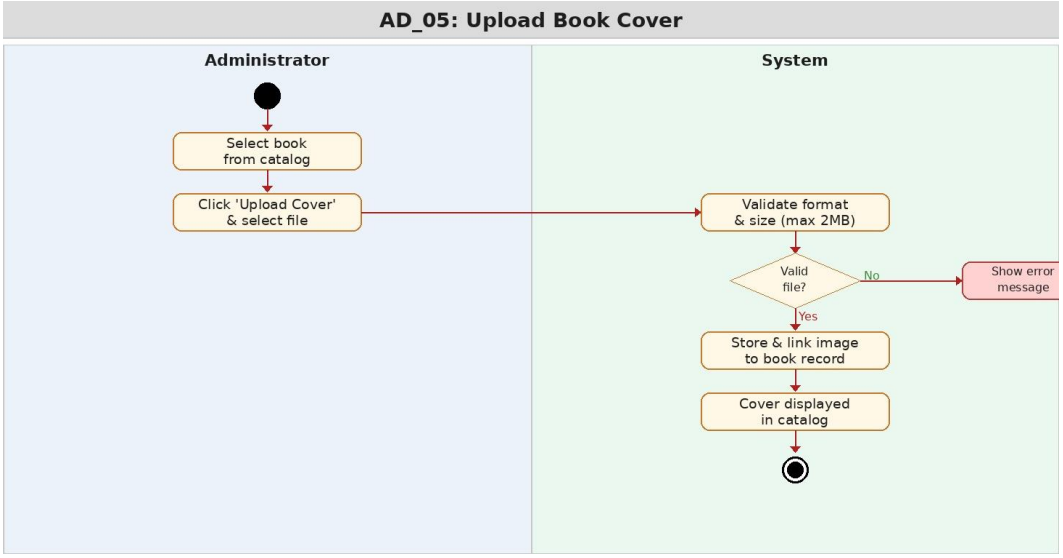
AD_03: Delete Book from Catalog



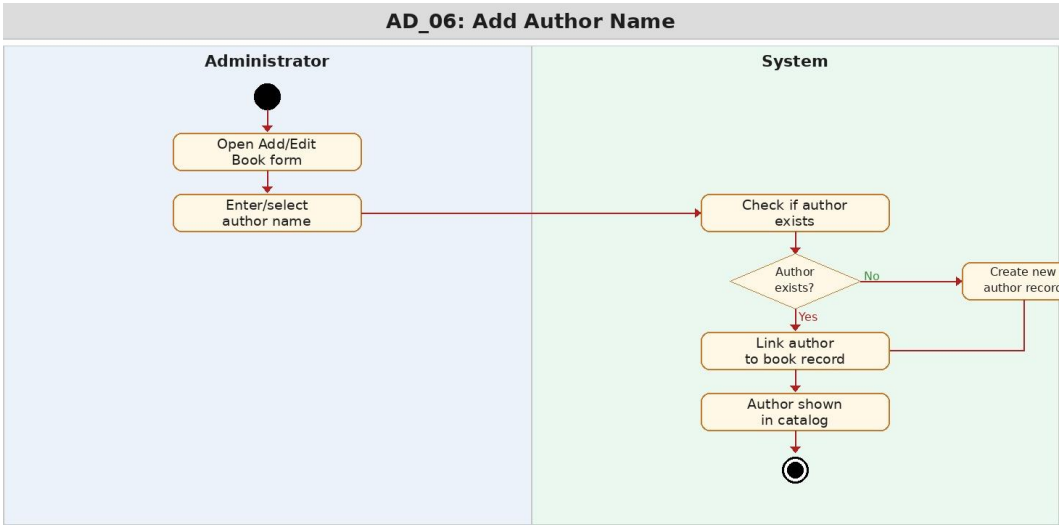
AD_04: View Book Details



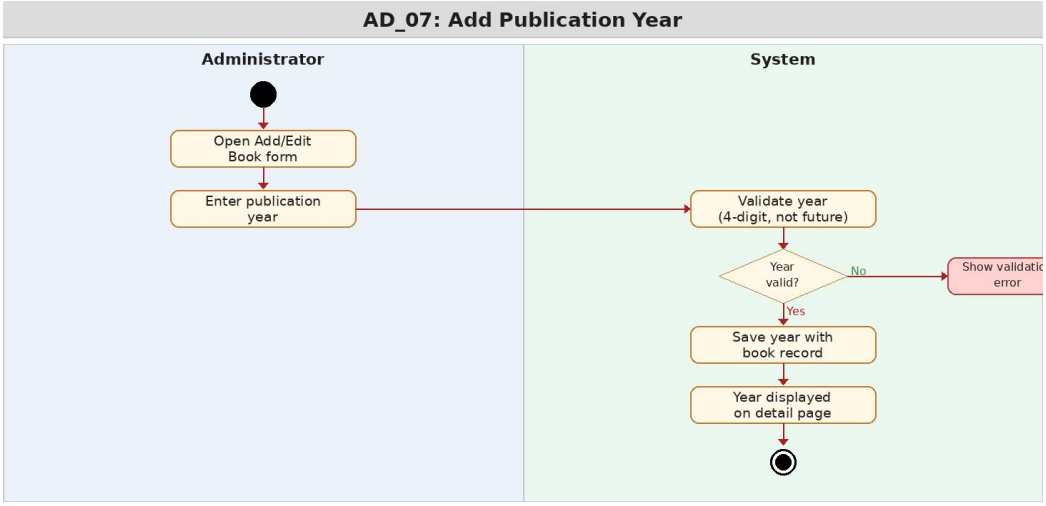
AD_05: Upload Book Cover



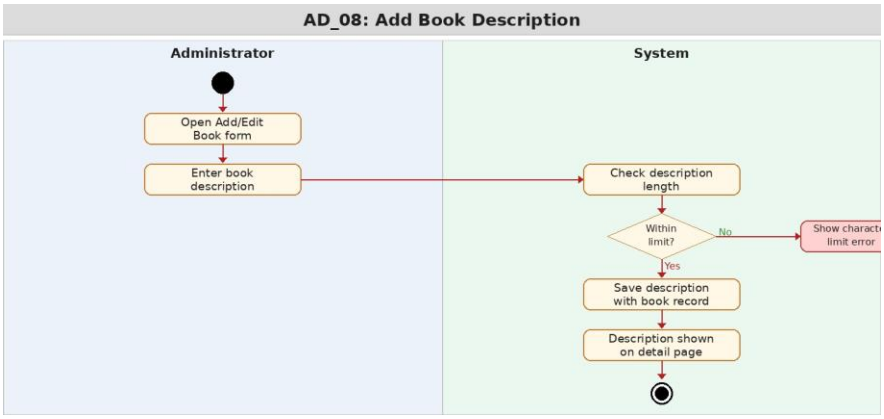
AD_06: Add Author Name



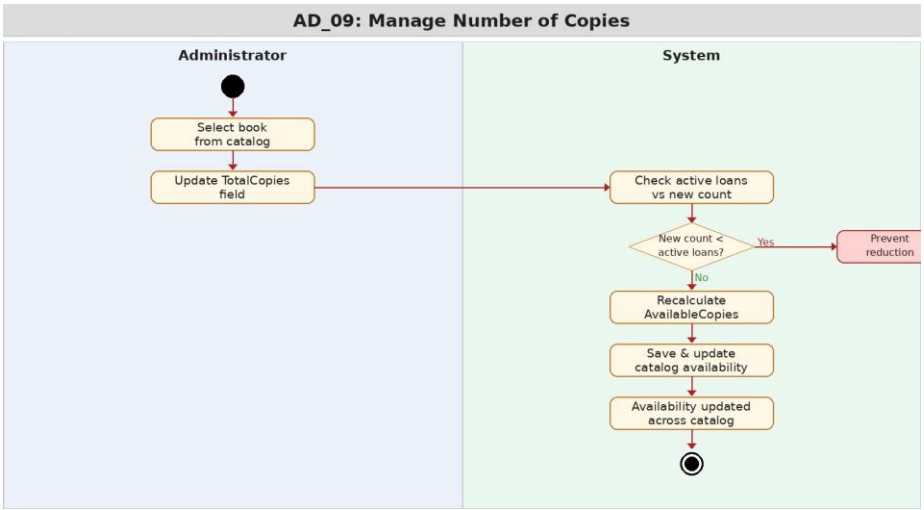
AD_07: Add Publication Year



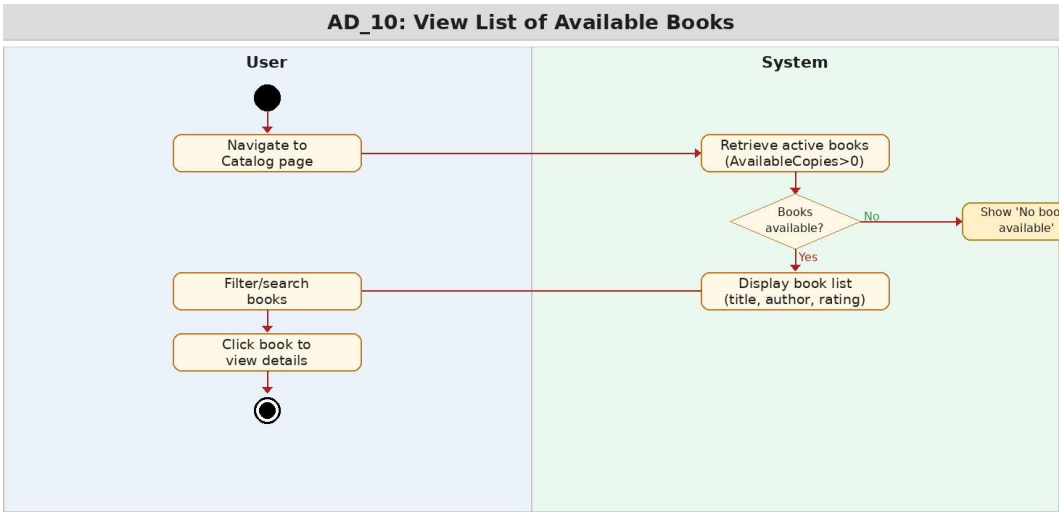
AD_08: Add Book Description



AD_09: Manage Number of Copies

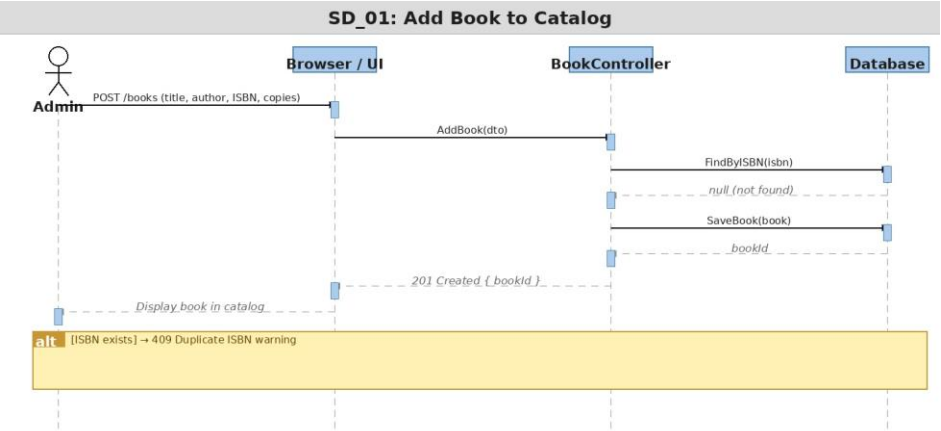


AD_10: View List of Available Books

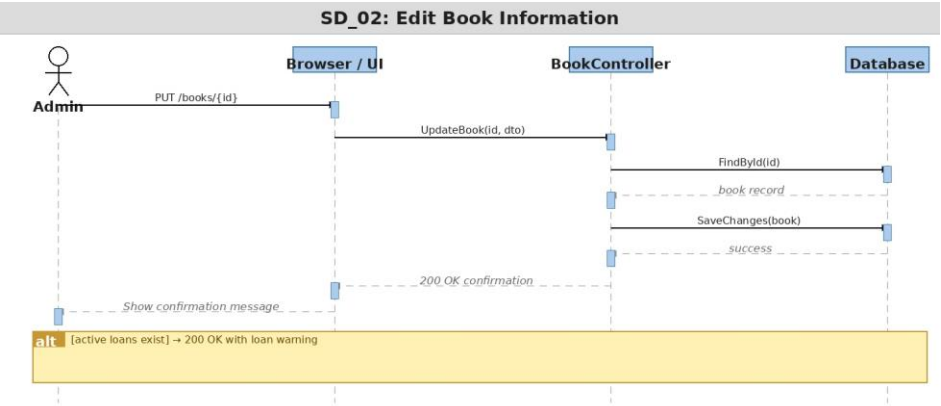


Sequence Diagrams

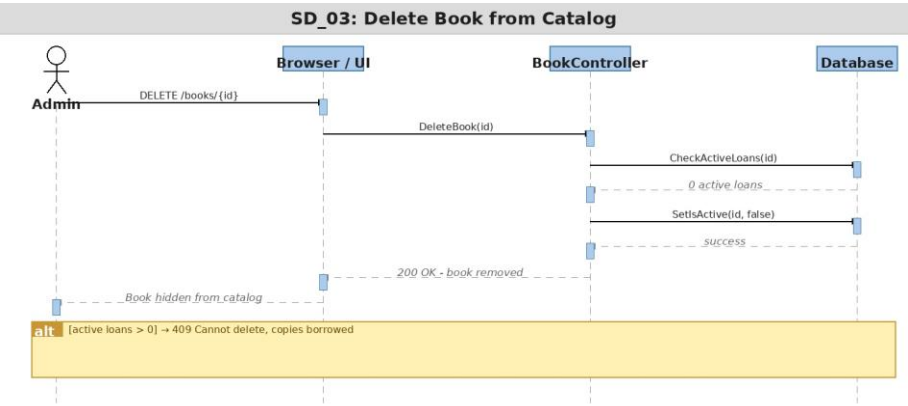
SD_01: Add Book to Catalog



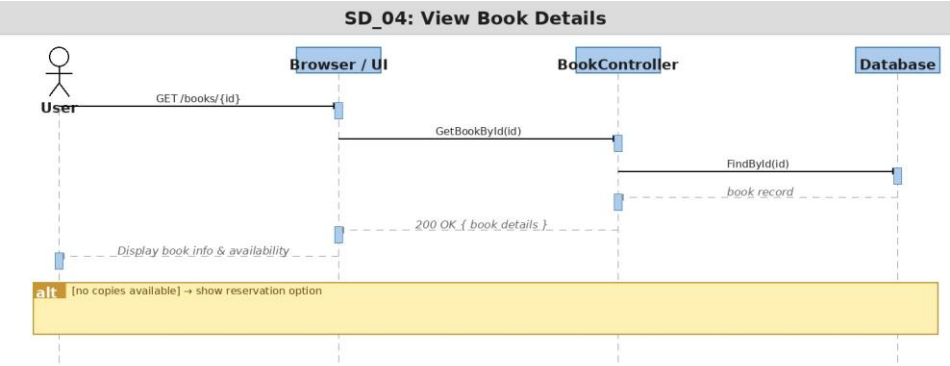
SD_02: Edit Book Information



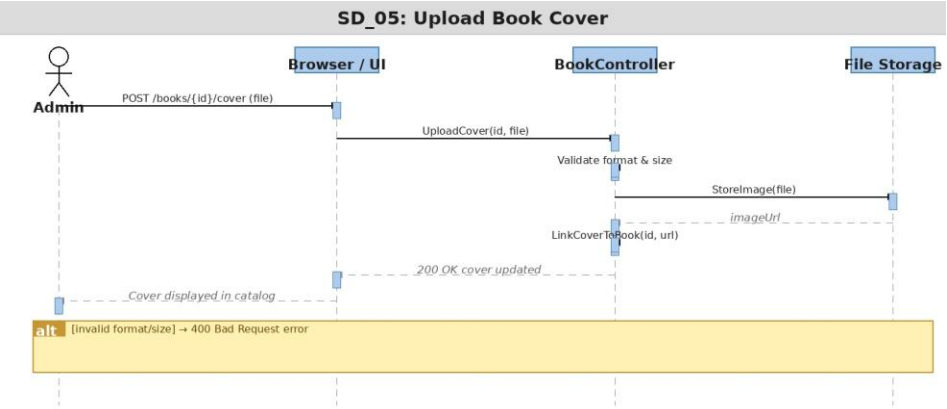
SD_03: Delete Book from Catalog



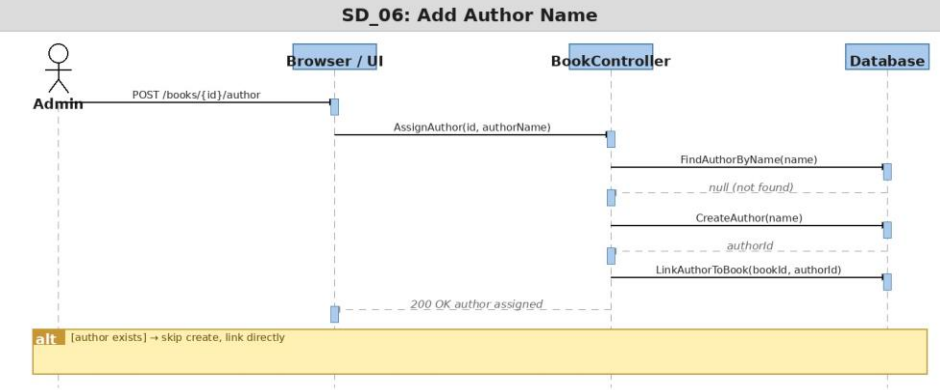
SD_04: View Book Details



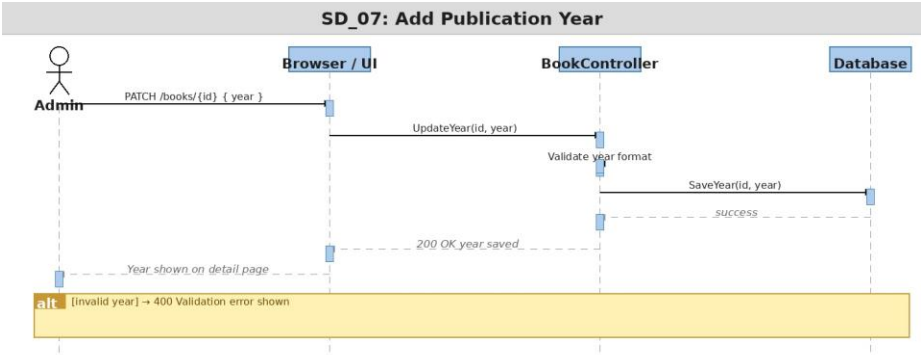
SD_05: Upload Book Cover



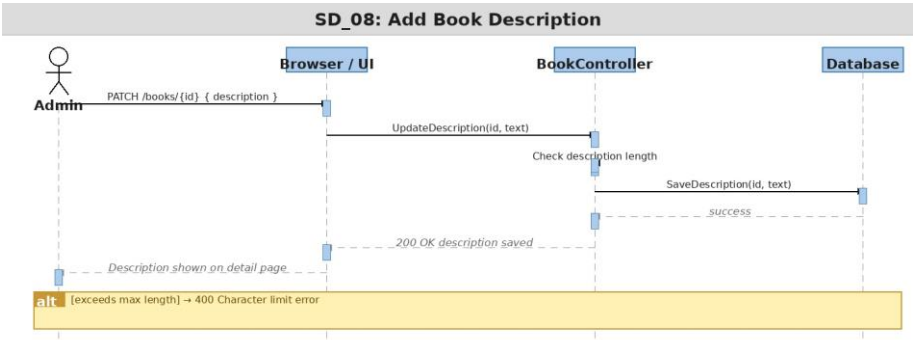
SD_06: Add Author Name



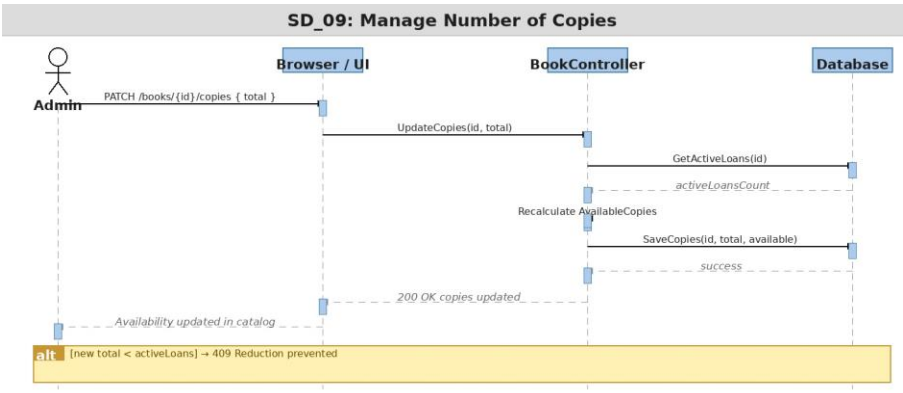
SD_07: Add Publication Year



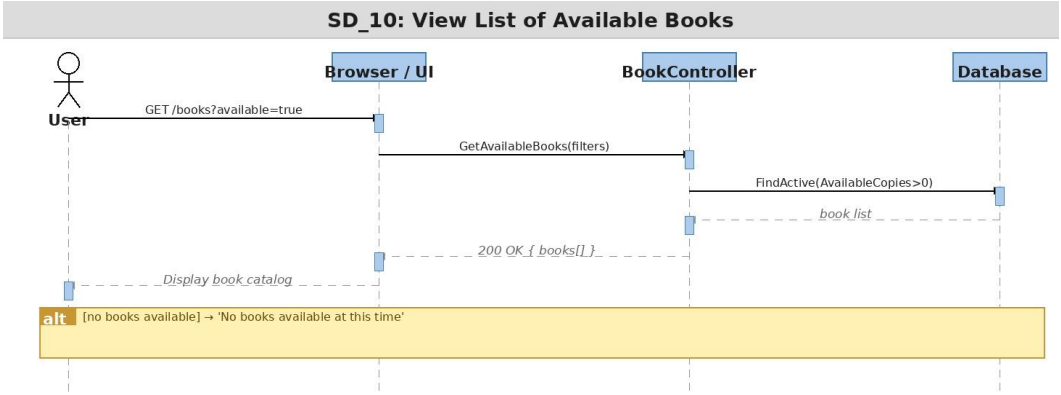
SD_08: Add Book Description



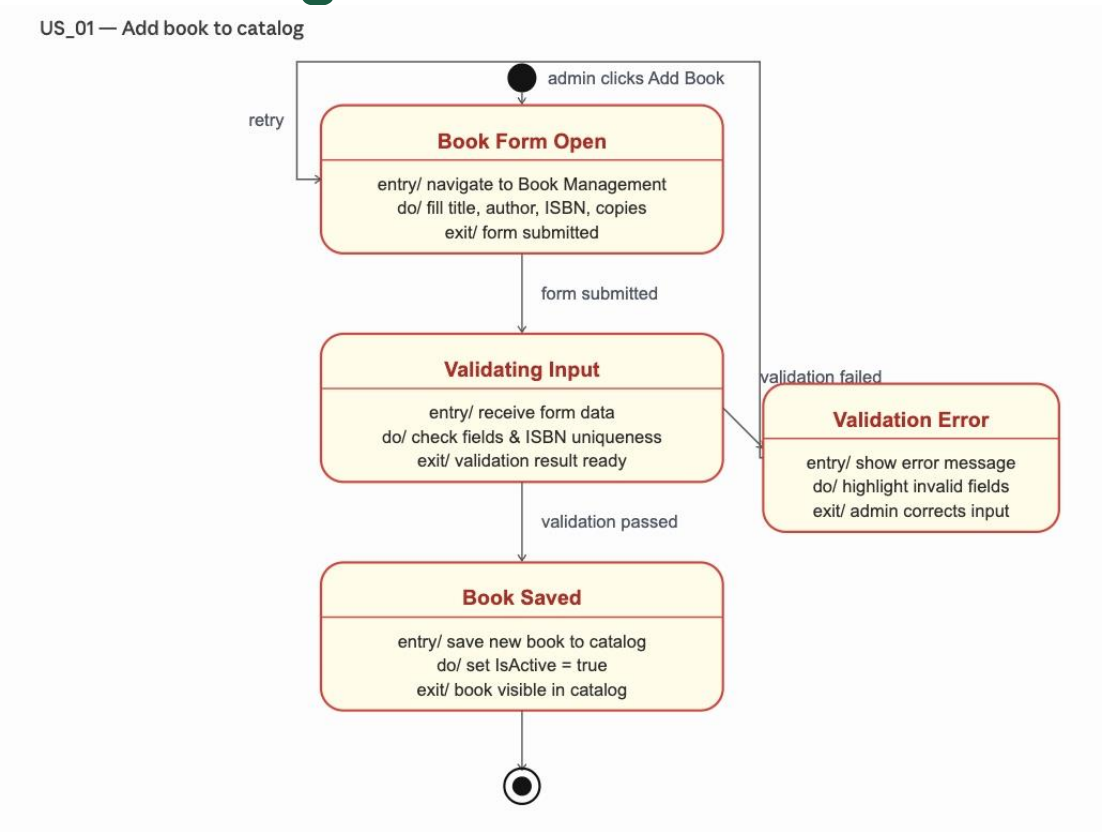
SD_09: Manage Number of Copies



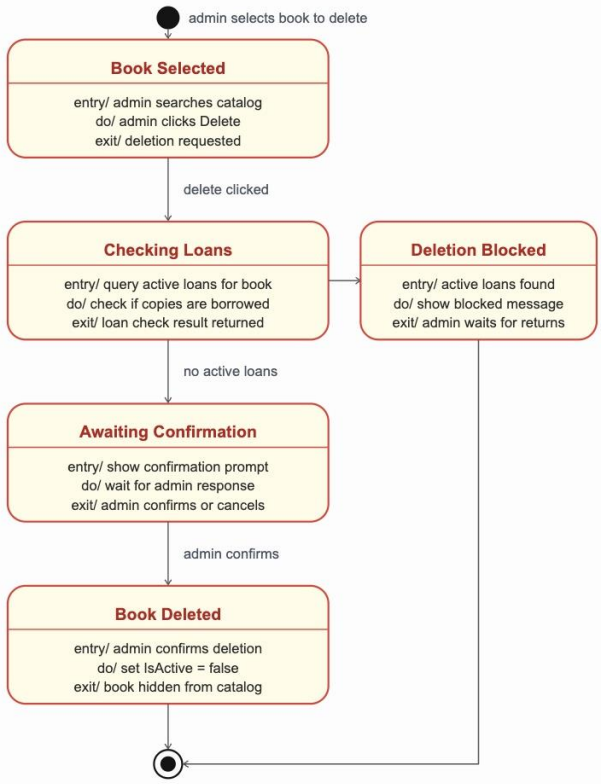
SD_10: View List of Available Books



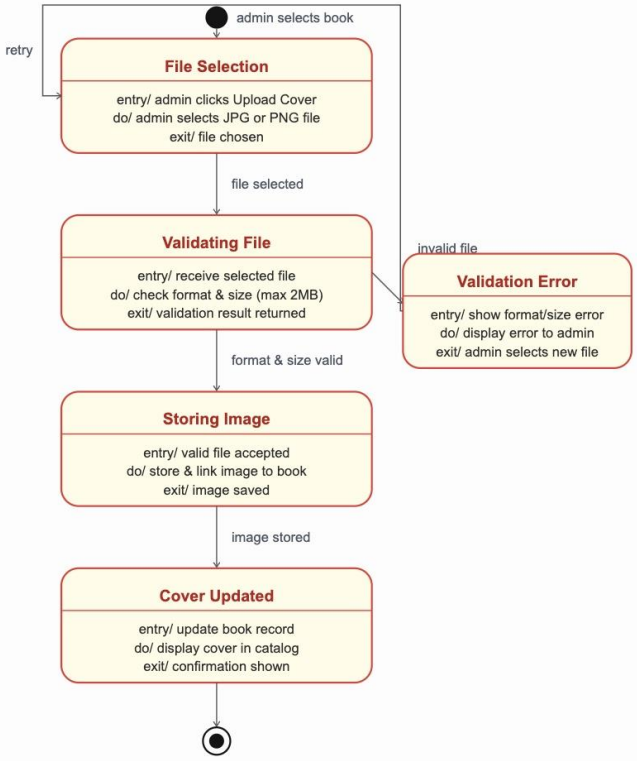
State Diagrams



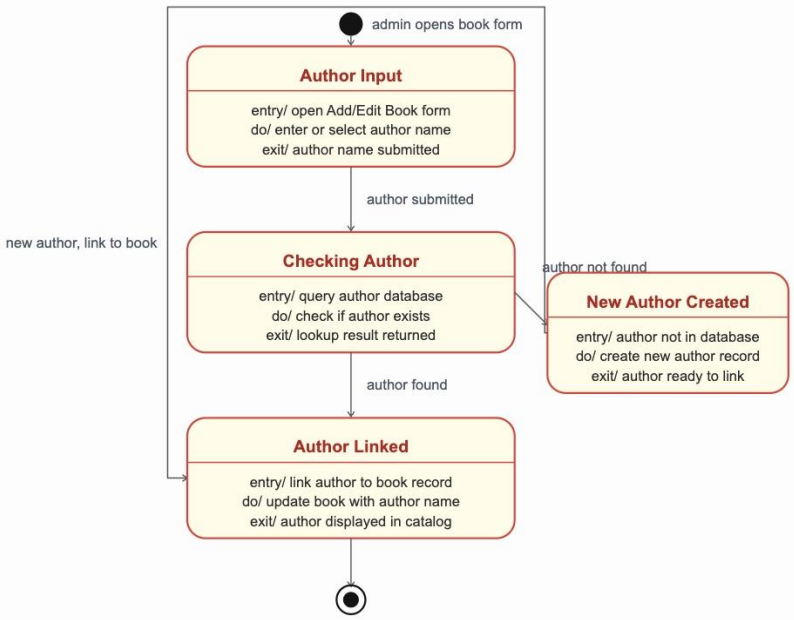
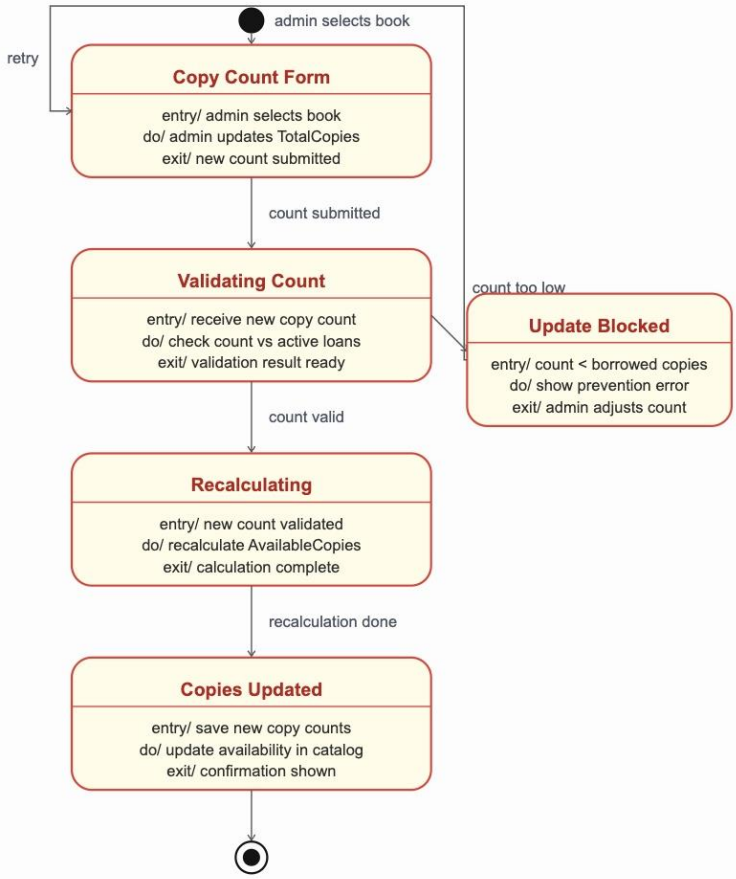
US_03 — Delete book from catalog



US_05 — Upload book cover



US_06 — Add author name



US_09 — Manage number of copies